Nanoscale

ROYAL SOCIETY OF CHEMISTRY

PAPER

Published on 05 September 2020
Licence and permissions

Check for updates

Cite this: *Nanoscale*, 2020, **12**, 19359

Me-graphene: a graphene allotrope with near zero Poisson's ratio, sizeable band gap, and high carrier mobility†

Zhiwen Zhuo, [a] Xiaojun Wu [a,b]* and Jinlong Yang [a,b]

The exploration of new two-dimensional (2D) allotropes of carbon has attracted great research attention after graphene, but experiment-feasible graphene allotropes with novel properties are still rare. Here, we predict a new allotrope of graphene, named Me-graphene, composed of both sp^2- and sp^3-hybridized carbon by topological assembly of C-$(C_3H_2)_4$ molecules. With a transitional ratio of sp^2- and sp^3-hybridized carbon atoms (12 : 1) between those of graphene (1 : 0) and penta-graphene (2 : 1), Me-graphene has transition properties between those of graphene and penta-graphene, such as energy, band gap, and Poisson's ratio. Unusually, Me-graphene exhibits a near zero Poisson's ratio of from −0.002 to 0.009 in the *xy*-plane (or called "anepirretic"), different from that of graphene (0.169) and penta-graphene (−0.068). More importantly, the near zero Poisson's ratio behavior remains in a large strain range, being less than ±0.02 for strain from −15% to +3%. Me-graphene possesses an indirect band gap of 2.04 eV, as a transition of graphene (semimetal) and penta-graphene (wide band gap), and turns into a direct-bandgap semiconductor with an enlarged band gap of 2.62 eV under compressive strain. It possesses high hole mobility of 1.60 × 10^5 cm^2 V^{-1} s^{-1} at 300 K. Me-Graphene has potential applications in electronic, photoelectric and high-speed mechatronic devices. The transitional properties related to the ratio of sp^2- and sp^3-hybridized carbon atoms are inspiring for searching for new graphene allotropes with combinational properties.

Received 19th May 2020,
Accepted 4th September 2020

DOI: 10.1039/d0nr03869e

rsc.li/nanoscale

Introduction

Carbon, the fourth richest element in the universe, presents various allotropic structures ranging from quasi-zero dimensional (0D) fullerene to three-dimensional (3D) graphite with versatile physical properties, partly due to its rich hybridized characteristics.[1] In recent years, graphene, a prototypical two-dimensional (2D) carbon allotrope with pure sp^2-hybridized carbon atoms packed into a honeycomb lattice, was first isolated in the laboratory in 2004, and has attracted intensive academic and industrial interest due to its intriguing properties, such as quantum Hall effect and ballistic electronic conductivity, thereby providing great opportunities in post silicon nanoelectronics and energy devices.[2–4] However, the absence of a sizeable bandgap in pristine graphene hinders its potential applications in electronics. Since the discovery of graphene, a large family of 2D graphene allotropes containing a mixture of sp, sp^2, and sp^3-hybridized carbon atoms has been reported theoretically and experimentally. In particular, the mixture of sp, sp^2, and sp^3-hybridized carbon atoms with different ratios endows 2D graphene allotropes with rich electronic properties ranging from metal,[5–7] to semimetal,[8–11] to semiconductor,[12,13] and various magnetic properties,[14–16] significantly extending the application of 2D graphene allotropes in nanoscale electronics,[17–20] spintronics,[14–16] catalysis,[21] thermal electronics,[22] energy storage,[6,23] and gas separation.[24] For instance, several semiconducting graphene allotropes, including graphdiyne (GD),[19] T-CA,[20] pentagonal carbon nanoflakes[18] and tetrahexcarbon,[17] have been theoretically predicted to possess high carrier mobility. Experimentally, graphdiyne with a mixture of sp and sp^2-hybridized carbon has been synthesized,[25] which can exhibit a high carrier mobility of 2.08 × 10^5 cm^2 V^{-1} s^{-1} and a narrow bandgap at room temperature.[19] To date, it is still of particular interest to explore new experimentally feasible 2D semiconducting carbon allotropes with intriguing properties.

Poisson's ratio (ν) is a fundamental property in the mechanical properties of materials, and represents the **d**eformation

[a]*Hefei National Laboratory for Physical Sciences at the Microscale, School of Chemistry and Materials Sciences, CAS Key Laboratory of Materials for Energy Conversion, and CAS Center for Excellence in Nanoscience, University of Science and Technology of China, Hefei, Anhui 230026, China. E-mail: xjwu@ustc.edu.cn*
[b]*Synergetic Innovation of Quantum Information & Quantum Technology, University of Science and Technology of China, Hefei, Anhui 230026, China*

†Electronic supplementary information (ESI) available. See DOI: 10.1039/d0nr03869e

response in **r**elaxable cross-**d**irection(s) (DRRD) to an orthogonal strain. It is generally affected by the structure geometry[26–28] as well as the electronic properties[29,30] and exhibits various mechanical responses in different ways with various amplification degrees under certain strain ranges.[31–36] In response, the DRRD can be meiotic[37] or auxetic, of which "meiotic" represents positive Poisson's ratio (PPR) and "auxetic" represents negative Poisson's ratio (NPR). In response amplification, there is an ambiguous zone between relatively significant PPR and NPR, known as "anepirretic", which has near zero Poisson's ratio (ZPR).[37] For response to strain, a material that exhibits a change of structure and electronic properties under strain consequently has a strain-dependent Poisson's ratio, which may vary from PPR to ZPR to NPR within different strain ranges.[30,38] Most materials naturally exhibit PPR mechanically as meiotic materials, while auxetic materials with NPR are relatively rarely found, and exhibit improved mechanical properties, such as enhanced shear moduli, indentation resistance and fracture toughness, as well as absorb vibrations and sound better than their non-auxetic counterparts.[31–33] Different to conventional PPR and NPR materials with technically large deformation responses, anepirretic materials have ignorable deformation responses to avoid strain effects in relaxable direction(s), with the advantages of reducing mechanical stress and shape change during compressive and tensile strain to be perfect in sealing, skins, and morphing aircraft applications for cellular structures,[39–41] and increasing the means for strain manipulation into positive, negative and null as responses and outputs. The well-known near ZPRs are found in gas (zero), cork (0 ± 0.05),[42] and spongy graphene ($\sim$0.01).[43] In 2D structures, there are also some materials with relatively small in-plane Poisson's ratio in partial directions, such as borophene (-0.02),[44] γ-2D silica (-0.022),[45] and Ag_2S (0.034),[46] except for α-phosphorene with an out-of-plane one of -0.027.[47] Though the Poisson's ratio of a material may be found to be zero in a few specific directions in geometry or be adjusted from a normal value into ZPR under strain,[46,48] the natural 2D anepirretic materials with highly near ZPR ($|\nu| < 0.01$) on average are quite rare. Structurally, the Poisson's ratio of materials can be adjusted by altering the repeated geometry unit.[28] With this idea, 2D anepirretic materials with near ZPR are likely able to be designed by geometry combination of structural units with the opposite deformation response to uniaxial strain, such as the combination of sp^3-C in penta-graphene (negative ν of -0.068)[13] and benzene rings in graphene (positive ν of 0.169)[49] in an appropriate ratio.

In this work, we propose a new graphene allotrope, named Me-graphene, by topologically assembling molecules of C-$(C_3H_2)_4$ with first-principles calculations. Consisting of sp^2 and sp^3-hybridized carbon atoms at the ratio of 12 to 1, Me-graphene is a transition between graphene and penta-graphene in structure as well as properties, including formation energy, band gap and Poisson's ratio. Me-graphene has a sizeable indirect band gap of 1.08 eV, comparable to that of silicon, and a high carrier mobility of 1.60×10^5 cm^2 V^{-1} s^{-1} under room temperature of 300 K, comparable to those of graphene and GD. In particular, the appropriate combination of sp^2 and sp^3-hybridized carbon structural units endows Me-graphene with a near zero negative Poisson's ratio of from -0.002 to 0.009 within varied direction and high Young's stiffness of 210 N m^{-1}. The axial Poisson's ratio remains small (less than ± 0.02) for strains from -15% to $+3\%$. Both phonon spectrum calculations and Born–Oppenheimer molecular dynamic (BOMD) simulations confirm the high structure stability of Me-graphene. The formation energy is much smaller than that of synthesized GD, implying the possibility of realizing the structure experimentally. Our results demonstrate the great potential of Me-graphene in nanoscale electronics, photoelectrics and high-speed mechatronics. The transitional properties related to the ratio of sp^2- and sp^3-hybridized carbon atoms are helpful for searching for new graphene allotropes with combinational properties.

Methods

All computations were performed by the density functional theory (DFT) method implemented in the Vienna *ab initio* simulation package (VASP).[50] The projector augmented wave (PAW) method[51] was applied to describe the electron–ion interaction with valence electrons on the carbon element in the configuration of $2s^2 2p^2$. The energy cut-off of the plane wave is 500 eV. The generalized gradient approximation (GGA) in the Perdew, Burke, and Ernzerhof (PBE) functional[52] was used as the exchange–correlation interaction functional. The Heyd, Scuseria, and Ernzerhof (HSE06)[53,54] screened hybrid density functional method and GW0[55] were employed to obtain more accurate electronic properties. Phonon calculation was carried out by the finite displacement method implemented in Phonopy code.[56] The convergence of force for optimizing the structure is 0.01 eV Å^{-1}, and 0.001 eV Å^{-1} for Poisson's response simulation. K-point sampling was performed in the Γ-center scheme of $16 \times 16 \times 1$ ($20 \times 20 \times 1$) to optimize the structure (calculate the single point energy and properties), respectively. The vacuum separation is 20 Å to avoid layered interaction. Born–Oppenheimer molecular dynamic simulation (BOMD) with the constant volume and constant temperature ensemble (NVT) was computed at the DFT level, of which the time step is 1 fs and the total simulation time is 5 ps for each temperature (from 500 to 4000 K with a temperature step of 500 K).

Results and discussion

Me-graphene from assembly of molecules

Our search for a new graphene allotrope began by topologically assembling the proposed molecule of C-$(C_3H_2)_4$, which contains twelve sp^2-hybridized and one sp^3-hybridized carbon atoms and can be considered as methane with four hydrogen atoms replaced with cyclocopropenylidenes,[57,58] as illustrated

 This journal is © The Royal Society of Chemistry 2020

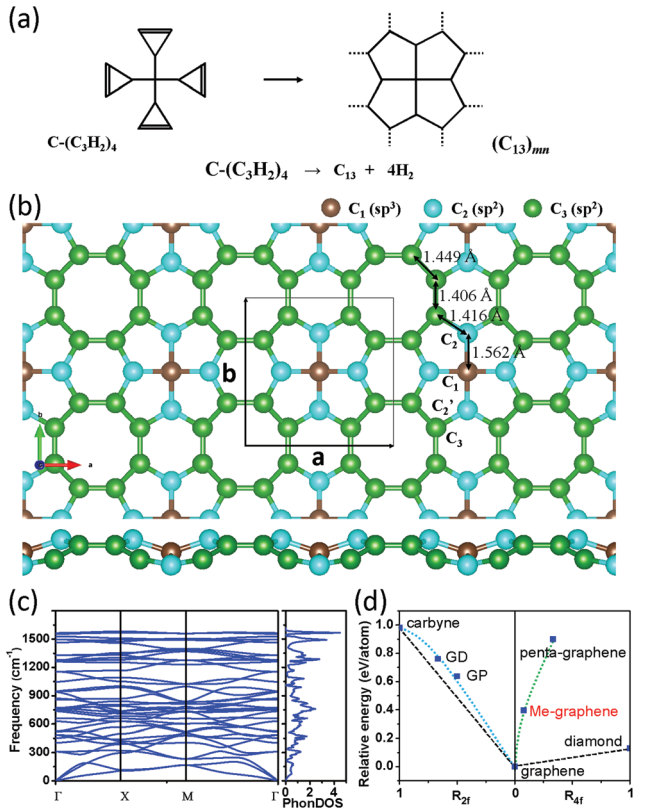


Fig. 1 (a) Me-graphene topologically assembled from the proposed molecule of C-$(C_3H_2)_4$. (b) The top view and side view of the Me-graphene structure with three kinds of equivalent carbon atom, labeled as C_1, C_2 and C_3 in the structure. (c) The phonon dispersion (band) and spectrum (DOS) of Me-graphene. (d) The energy distribution of different allotropes with different ratios of $sp : sp^2 : sp^3$, of which the R_{2f} is defined as $N_{sp}/(N_{sp} + N_{sp^2})$, and R_{4f} is defined as $N_{sp^3}/(N_{sp^3} + N_{sp^2})$, where N_{sp}, N_{sp^2} and N_{sp^3} are the number of sp, sp^2 and sp^3 carbon atoms in the unit cell, respectively. The energy of graphene is set to zero. The black dashed lines are the ideal energy lines of the structure consisting of sp–sp^2 and sp^3–sp^2 carbon atoms, while the colored dashed lines are the real profile for the listed structures. GP represents graphyne.[59]

in Fig. 1a. This new graphene allotrope, named as "tertiary-methyl"-graphene (Me-graphene), with a symmetry of $P\bar{4}m2$ (space group no. 115), contains carbon pentagons, hexagons, and octagons, as shown in Fig. 1b. The optimized lattice constants of Me-graphene are $a = b = 5.744$ Å with a thickness of 0.988 Å, of which the bonding of the sp^3-hybrdized carbon atom (C_1) and its four neighboring benzene rings (sp^2-hybridized carbon atoms, C_2) leads to the conversion of the symmetry from standard T_d to D_{2d} in central carbon atom C_1 to result in smaller thickness. The bond length of C_1–C_2 is 1.562 Å, and the bond angle among C_2–C_1–C'_2 is 143.15 degrees, much larger than the standard sp^3 hybridization angle of 109.47 degrees. The remaining eight sp^2-hybridized carbon atoms (C_3) in the unit cell form octagons. There are three kinds of bond length between the sp^2-hybridized atoms of 1.406, 1.416 and 1.449 Å, respectively, close to that of the C–C bond in graphene (1.426 Å) and far from that in diamond (1.548 Å), suggesting that there is a large delocalized π bond in the structure, which can be confirmed by the distribution of charge deformation density on the carbon framework, as shown in Fig. S1 (see ESI†). More details of the structure information are displayed in Fig. S2 in the ESI.† Besides, the electron localization function (ELF) indicates that Me-graphene is highly covalent as shown in Fig. S3 in the ESI.†

Structure stability

The structure stability of Me-graphene was examined by lattice dynamics and molecular dynamics. Fig. 1c displays the calculated phonon spectrum of Me-graphene. The absence of an imaginary frequency in the entire Brillouin zone confirms that Me-graphene is stable in lattice dynamics. The phonon density of states (PhonDOS) shows that the state distribution is almost continuous over all of the distributed zone, different to that of penta-graphene with a large gap.[13] The highest vibration frequency in the phonon spectrum reaches as high as 1582 cm^{-1}, smaller than that of penta-graphene (1630 cm^{-1}, contributed by double bonds in the sp^2 hybridized carbon dimer), implying the outstanding character of a large delocalized π bond in the Me-graphene lattice, compared to penta-graphene. Next, the thermal stability of the Me-graphene lattice was checked with BOMD simulations with the NVT ensemble at the temperature ranges from 500 to 4000 K. The total simulation time is 5 ps and the time step is 1 fs. As shown in Fig. S4 (see ESI†), Me-graphene retains its structure at the temperature of 3500 K, but reconstructs at the temperature of 4000 K, implying ultra-high thermodynamic stability of Me-graphene.

Energy stability

Total energy calculations were performed to investigate the structural stability of Me-graphene. Fig. 1d displays the relative stability of Me-graphene compared with other 2D graphene allotropes. Here, the relative formation energy of graphene is set to zero. Purely considering the ratio of carbon atoms in different hybridizations of $sp : sp^2 : sp^3$ to estimate the energy assuming that the energy of a carbon atom in sp, sp^2 and sp^3 is equal to carbyne, graphene and diamond, respectively, the energy contribution to the structure is in the sequence of $sp^2 > sp^3 > sp$, so a higher ratio of sp^2 in the structure can lead to a lower formation energy for the system, as the black dashed lines show in Fig. 1d. The calculated formation energy of Me-graphene ($sp^2 : sp^3 = 12 : 1$) between that of graphene and penta-graphene, being higher than that of graphene ($sp^2 : sp^3 = 1 : 0$) of 0.397 eV per atom, but lower than that of penta-graphene ($sp^2 : sp^3 = 2 : 1$) of 0.504 eV per atom and other sp–sp^2 hybridization allotropes,[59] including GD ($sp : sp^2 = 2 : 1$) and graphyne ($sp : sp^2 = 1 : 1$) of about 0.364 and 0.241 eV per atom, respectively. Remarkably, for 2D structures consisting of sp^2 and sp^3, their energy change (to the ratio change of sp^2 hybridization carbon in structure) is much sharper than those of 2D structures consisting of sp and sp^2, where the energy of R_{4f} of 1/3 (penta-graphene) is close to that of R_{2f} of 1 (carbyne[60]). For the sp^3 carbon atom, there is a large in-plane tensile force to form sp^3 hybridization in a few atom thick 2D space, while there is not for the sp carbon atom. Thus, to

design or obtain low energy graphene allotropes consisting of sp^2 and sp or sp^3, a higher ratio of sp^2 should be better. Note that GD has been realized in experiments, and penta-graphene has been proposed to be synthesized from T12-carbon phase.[13] The calculated formation energy implies that Me-graphene is experimentally feasible from the viewpoint of formation energy.

Mechanical properties

The mechanical stability criterion is $C_{11}C_{22} - C_{12}^2 > 0$ and $C_{66} > 0$.[61] And the calculated result is $C_{11} = C_{22} = 210$ N m^{-1}, $C_{12} = -0.4$ N m^{-1} and $C_{66} = 103$ N m^{-1}. Thus, Me-graphene is mechanically stable. By using the equation to calculate the Young stiffness of isotropic 2D structures of $Y_{11} = Y_{22} = (C_{11}C_{22} - C_{12}^2)/C_{11}$,[61] the Y_{11} of Me-graphene is 210 N m^{-1}. This is slightly smaller than that of penta-graphene (268 N m^{-1} (ref. 13)), but much larger than that of GD (about 100 N m^{-1} (ref. 62)). The Poisson's ratio is calculated as $\nu = C_{12}/C_{11} = -0.002$, quite near zero, as a transitional value between that of graphene (0.169) and penta-graphene (−0.068),[13] suggesting a sign that a higher ration of sp^3 hybridized carbon in the atom ratio is better for finding graphene allotropes with ZPR, PPR and NPR. Further considering that the Poisson's ratio depends on direction, Me-graphene safely behaves as an anepirretic material in all directions in the x–y plane, and is quite isotropic with a near ZPR varying from −0.002 to 0.009 (directional average of about 0.004), indicating that the Poisson's ratio is absolutely zero for some specific directions, as shown in Fig. S5 in the ESI.† It should be noted that, since there are PPR and NPR in Me-graphene, it also can be technically regarded as a meiotic as well as auxetic material, depending on the application direction and range of strain, as well as the detection sensitivity.

To investigate the mechanical behavior more specifically, the response of one axis (b) under uniaxial strain from −15% to +5% in another axis (a) is determined, as shown in Fig. 2. The energy change of Me-graphene appears like a parabola with the maximum at −15% of +0.340 eV per atom (and +0.040 eV per atom at +5%), as shown in Fig. 2a. The strain dependent Poisson's ratio ν of axis y (or b) to axis x (or a) is defined as $\nu_{xy}(\sigma_x) = -\Delta\sigma_y(\sigma_x)/\Delta\sigma_x = -(\sigma_y(\sigma_x + \Delta\sigma_x) - \sigma_y(\sigma_x - \Delta\sigma_x))/(2\Delta\sigma_x)$, where σ and $\Delta\sigma_x$ are the strain value and the strain step in axis a (of 1% here); $\sigma_y(\sigma + \Delta\sigma)$ and $\sigma_y(\sigma - \Delta\sigma)$ are the strain of axis y at strain of axis x of $\sigma + \Delta\sigma$ and $\sigma - \Delta\sigma$, respectively. Interestingly, the structure exhibits NPR behavior at strain ranging from −13% to 0%, of which the max value of NPR is −0.019, about ten times that at 0% (−0.002), and the Poisson's ratio is within ±0.01 for strain from −1% to +1%, less than ±0.02 for strain from −15% to +3%, as shown in Fig. 2b. Consequently, the lattice b exhibits itself a quite tiny change, less than 0.17%, equal to 0.01 Å, indicating that Me-graphene is quite uniaxial-strain-insensitive, as a ZPR material (highly anepirretic material), which behaves differently from PPR and NPR materials to achieve three different deformation responses of positive, negative and null for strain manipulation, as shown in Fig. 2c. Technically, if the ZPR is defined as $|\nu| < 0.01$, the Me-graphene performs as a ZPR material for full in-plane directions.

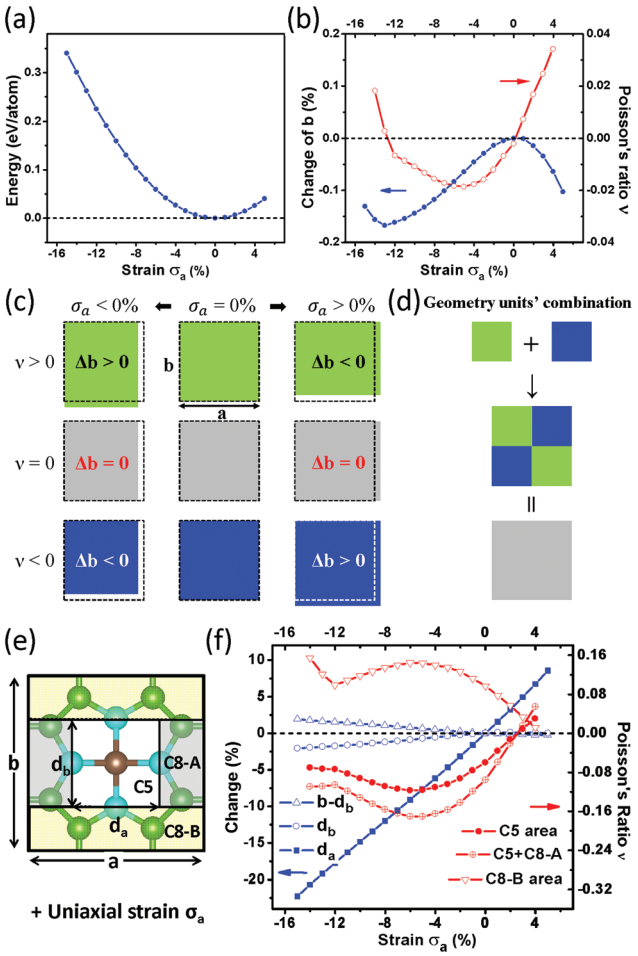


Fig. 2 (a) Energy change under uniaxial strain. (b) The changes of axis b and Poisson's ratio under uniaxial strain. (c) The different deformation response of PPR, ZPR and NPR materials. (d) Illustration of the ZPR unit geometrically constructed by NPR and PPR units. (e) The illustration of different zones and distances in Me-graphene. (Clean zone is denoted as C5 area while the left color zone is denoted as C8 area, of which there are two parts: A (in gray) and B (in yellow).) (f) The response of the distances and defined Poisson's ratio in (e) under uniaxial strain applied on axis a.

Assuming the geometric units' combination leads to units' mechanical response combination, the ZPR can be constructed by the auxetic area (or NPR part) and meiotic area (or PPR part) in geometry, as shown in Fig. 2d. In Me-graphene, the response of one axis (b) can be regarded as the combination of two opposite response areas, the auxetic area (denoted as C5 area, as shown in Fig. 2e) and the meiotic area (C8 area, shown in gray (part A) and yellow colors (part B)). For the auxetic area, the novel behavior of sp^3-hybridized carbon atoms is the key to introduce NPR into the structure, just like sp^3-C in penta-graphene[13] and sp^3-Si in 2D SiO_2 with NPR,[45] when the strain is applied on direction in the D_{2d} plane of the sp^3-hybridized tetrahedron of SiO_4. Meanwhile, the C8 area consists of benzene rings as the meiotic area, which perform

This journal is © The Royal Society of Chemistry 2020

like graphene with a positive Poisson's ratio. To investigate the response of different areas, the structure can be divided in two parts in two ways, separated C5 area and crossed C8 area as the first way, and two parallel areas: C5 + C8-A area and C8-B area for the other way, of which the length rate of the C5 area as well as the C5 + C8-A area in the initial structure is $d_{b,\ \sigma\ =\ 0}/b_{\sigma\ =\ 0}$, about 51.6%, slightly larger than that of the C8-B area. Based on the change of lengths of different areas, as shown in Fig. 2f, the largest NPR of the C5 + C8-A area ($-\Delta d_b/\Delta a$) is calculated to be −0.171, while the largest Poisson's ratio of the C8-B area ($-\Delta(b - d_b)/\Delta a$) is 0.154. And the largest apparent NPR of the C5 part ($-\Delta d_b/\Delta d_a$), which is the combination response of the C5 area and C8-A area in the structure, is −0.117, smaller than that of C5 + C8-A, because the strength of sp^3-C is lower than that of sp^2-C, leading to a larger change degree in a series-connected structure under strain. The responses of the auxetic area and the meiotic area are opposite for most of the strain range (−15% to +3%), of which the C5 area turns into a meiotic one when the strain is larger than +3%. They cancel out so well that the Poisson's ratio is almost zero, ranging from −0.019 to 0.034, and only −0.002 at a strain of 0% to make the Me-graphene uniaxial-strain mechanical response insensitive, different from the high ratio of sp^3-C leading to obvious negative Poisson's ratio in penta-graphene.

Electronic properties

Me-graphene is an indirect band gap semiconductor with a band gap of 0.65 eV with PBE (Γ (0, 0, 0) → M (0.5, 0.5, 0)), as shown as Fig. 3a. Partial density of states (PDOS) suggests that the main state around the Fermi level is contributed by p_x and p_z of C_2 and p_z of C_3, of which p_z of C_2 is the main contributor near the conduction band minima (CBM) as well as the valence band maximum (VBM), as shown in Fig. 3c & d, suggesting that C_2 is in a higher active state than C_3, in agreement with the situation of C_2 with a higher structure stress. The partial charge density distributions of the CBM and VBM are shown in Fig. 3b, of which both of them all locate on the octagonal rings, but with spatial separation. As is known, PBE always underestimates the band gap, and the hybridization function of the HSE06 method was employed to obtain a more accurate band gap. The band gap increases to 1.08 eV after calculation in HSE06. Besides, the energy levels (*vs.* vacuum energy level, set to 0 eV) of the VBM and CBM as well as the work function in HSE06 are −5.09, −4.01 and 5.09 eV, respectively. It should be noted that many studies have shown that, for 2D systems, the HSE06 functional tends to underestimate the intrinsic band gap to a degree. Thus, a GW0 method is applied to obtain a more accurate intrinsic band gap. The quasiparticle band gap is estimated to be 2.04 eV with GW0 (see ESI† for more calculation details), larger than that of HSE06 by 0.96 eV. For the band gap feature in 2D graphene allotropes, there is a trend that a higher ratio of sp^3 to sp^2 can lead to a larger band gap, where the band gap is in the sequence of graphene (R_{4f} = 0, 0 eV) < Me-graphene (R_{4f} = 12/13, 2.04 eV) < penta-graphene (R_{4f} = 1/3, 4.48 eV (ref. 63)) < diamond (R_{4f} = 1, 5.5 eV (ref. 64)), because the sp^3 can localize the electronic states to the fragment delocalized π bond to open the band gap. Considering both the energy stability and band gap opening degree of the Me-graphene, and the transition between graphene and penta-graphene, the balance ratio of sp^2 and sp^3 carbon atoms needs to be adjusted in different design requirements.

The change of structure and properties under biaxial strain ranging from −5% to +5% was investigated, as shown in Fig. 4, S7, S8, and Table S2.† The energy change of the structure is small, of which the maximum for the tensile and compressive strain is +0.085 eV per atom at +5% and +0.082 eV per atom at −5%, respectively, as shown in Fig. 4a. There is a direct-indirect band gap transition between −3% and −2%, of which the

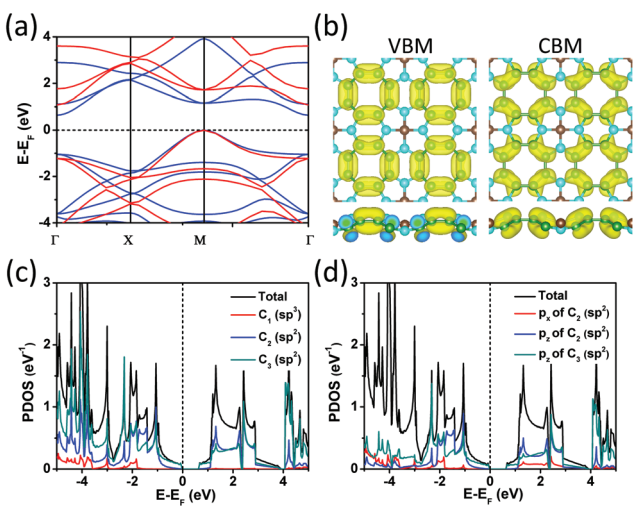


Fig. 3 (a) Band structure of Me-graphene with PBE (blue) and HSE06 (red). (b) Isosurface (yellow color, 0.005 e Bohr^{-3}) of partial charge densities of the VBM and CBM. (c) The PDOS from different contributed atoms. (d) The PDOS of the main contributed orbitals.

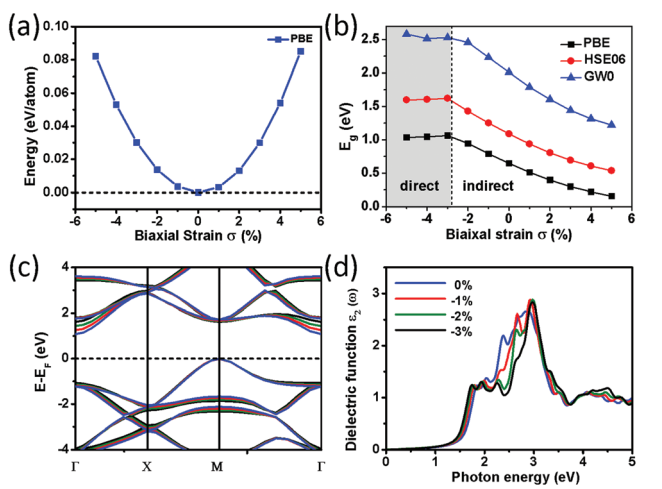


Fig. 4 (a) Energy change and (b) band gap change under biaxial strain from −5% to +5%. (c) Band structure change (HSE06) and (d) the imaginary part of the dielectric function under strain from 0% to −3%, plotted with different colors by HSE06.

band gap is direct type (M → M) from −5% to −3% while it is indirect type (Γ → M) from −2% to +5%, as shown in Fig. 4b & c. This results from the strain-sensitive energy shift of the lowest unoccupied state (LUS) at Γ, due to the sharpest change in the p_z–p_z overlap leading to weak coupling at compressive strain and stronger coupling at tensile strain (see ESI† for more details). In general, the band gap changes from 2.62/1.60 (at −5%) to 1.24/0.54 eV (at +5%) with GW0/HSE06, and a similar change trend is found with PBE.

For optical properties, the general PBE based method was carried out without the exciton effect. Based on the highly linear relationship between exciton binding energy (E_b) and GW0 band gap unveiled in previous works,[65,66] with the fitting formula of $E_b = 0.21E_g + 0.40$, the E_b can be conveniently estimated when the GW0 band gap is known. For Me-graphene with a GW0 band gap of 2.04 eV, the exciton binding energy is calculated to be 0.83 eV by the fitting formula, and the optical band gap is estimated to be 1.21 eV, quite close to that with HSE06 (1.08 eV). It is similar to that of phosphorene, which has a quasiparticle band gap of 2.2 eV (*vs.* optical band gap of about 1.3 eV) with an exciton binding energy of about 0.9 eV in experiments.[67] Therefore, the HSE06 method is employed as a substitution of other high cost methods for optical properties. This result shows the imaginary part of the dielectric function, indicating that the absorption range of photons covers infrared, visible and ultraviolet light with high intensity. During the strain change from 0% to −3%, the optical absorption edge changes slightly but with an obvious change in the absorption peaks in the visible light zone, as shown as Fig. 4d. Still the value of optical absorption significantly remains over a large energy range. In general, the optical property of Me-graphene possesses excellent optical absorption covering a wide energy range at strain from 0% to −3%, which can be applied as a photoactive donor material in solar cells, and other photoelectric devices.

Carrier mobility

Based on the deformation potential theory proposed by Bardeen and Shockley[68] to calculate carrier mobility, which has been successfully applied in single-element materials in previous works,[69,70] we have investigated the carrier mobility of Me-graphene. The effective mass of Me-graphene at the CBM and VBM in the axial a (as well as b) direction ($m^*_{VBM,a}$ and $m^*_{CBM,a}$) is −0.18 and $0.68m_e$, respectively. The deformation potential $E_{1,VBM,a}$ and $E_{1,CBM,a}$ is −0.93 and −7.86 eV, respectively. At room temperature (300 K), the calculated carrier mobilities of μ_h and μ_e in direction a are 1.60×10^5 and 157 $cm^2\ V^{-1}\ s^{-1}$, respectively. The dramatic difference between the mobility of electrons and holes (1/1019) results from the significantly small value of effective mass (0.18, *vs.* 0.68) as well as deformation potential (0.93, *vs.* 7.86) of the hole carrier (*vs.* electron carrier) (see ESI† for more details). In comparison, the hole carrier mobility μ_h of Me-graphene in the a direction is roughly comparable to those of graphene (3.39×10^5 $cm^2\ V^{-1}\ s^{-1}$)[71] and graphdiyne (2.08×10^5 $cm^2\ V^{-1}\ s^{-1}$).[19] It is also much better than that of phosphorene ($1–2 \times 10^4$ $cm^2\ V^{-1}\ s^{-1}$ in theory prediction),[70] which has been demonstrated to reach 5×10^3 in experiments.[72] Moreover, Me-graphene possesses a band gap of 2.04 eV, much larger than that of silicon (1.1 eV),[73] and comparable to that of phosphorene (2.2 eV),[67] indicating its potential promising application in electronic devices.

Layered vdW structure

Two vdW layered bulks, AA stacking and AB stacking, are investigated by the DFT + D3 method.[74] The optimized lattice parameters of the AA stacking vdW layered bulk denoted as *P*$\bar{4}$*m2-lb*-C_{13} (space group no. 115) are $a = b = 5.749$ Å, $c = 3.698$ Å, as shown in Fig. S11a.† Those of the AB stacking structure denoted as *AMM2-lb*-C_{26} (space group no. 38) are $a = 5.752$, $b = 5.753$ Å, $c = 7.926$ Å, as shown in Fig. S5b.† The smallest interlayer distance of each stacking is 3.479 and 3.308 Å, respectively. The interlayer interaction is about −39 meV per atom for AA stacking and −33 meV per atom for AB stacking, suggesting that the AA stacking is more stable than AB stacking in energy. The band gap of the AA and AB stacking bulks is 0.76/0.15 and 0.96/0.33 eV, much smaller than that of the monolayer by 1.28/0.51 and 1.08/0.33 eV with different methods (GW0-D3/DFT+D3), respectively, suggesting that the vdW interactions can significantly reduce the band gap to adjust it to a certain smaller band gap, as shown in Fig. S11c & d.†

Conclusion

A new graphene allotrope with a novel structure consisting of sp^2- and sp^3-hybridized carbon atoms is proposed here. Its stability is confirmed by the phonon spectrum and Born–Oppenheimer molecular simulation. Me-graphene is a transition between graphene and penta-graphene in structure and many properties. It naturally has a near zero Poisson's ratio of from −0.002 to 0.009, which remains small (less than ±0.02) for strain from −15% to +3% in the axial direction, and a high Young's stiffness of 210 N m^{-1}, acting as a ZPR material (anepirretic material) as well as an auxetic material. The structure has an indirect band gap of 2.04 eV, which can be adjusted into a direct band gap up to 2.62 eV under compressive strain. It possesses excellent photon absorption covered infrared, visible and ultraviolet light. Furthermore, it possesses high carrier mobility (1.60×10^5 $cm^2\ V^{-1}\ s^{-1}$) at room temperature (300 K). These interesting, inspiring and transitional results indicate that Me-graphene is a promising material in electronic, photoelectric and high-speed mechatronic devices in the nanoscale. The transitional properties related to the ratio of sp^2- and sp^3-hybridized carbon atoms reveal a new door to searching for new graphene allotropes with combinational properties to be multi-functional materials.

Conflicts of interest

There are no conflicts to declare.

This journal is © The Royal Society of Chemistry 2020

Acknowledgements

This work was supported by the MOST (2018YFA0208603, 2016YFA0200602), NSFC (11574282, 21573204, 21890751), Strategic Priority Research Program of CAS (XDB01020300), the Anhui Initiative in Quantum Information Technologies, the Fundamental Research Funds for the Central Universities, National Program for Support of Top-notch Young Professional, External Cooperation Program of BIC CAS (211134KYSB20130017), and the Super Computer Centre of USTCSCC and SCCAS.

References

1 Z. Zhao, F. Tian, X. Dong, Q. Li, Q. Wang, H. Wang, X. Zhong, B. Xu, D. Yu, J. He, H.-T. Wang, Y. Ma and Y. Tian, *J. Am. Chem. Soc.*, 2012, **134**, 12362–12365.
2 K. S. Novoselov, A. K. Geim, S. V. Morozov, D. Jiang, Y. Zhang, S. V. Dubonos, I. V. Grigorieva and A. A. Firsov, *Science*, 2004, **306**, 666–669.
3 K. S. Novoselov, A. K. Geim, S. V. Morozov, D. Jiang, M. I. Katsnelson, I. V. Grigorieva, S. V. Dubonos and A. A. Firsov, *Nature*, 2005, **438**, 197–200.
4 A. K. Geim, *Science*, 2009, **324**, 1530.
5 H. Terrones, M. Terrones, E. Hernández, N. Grobert, J. C. Charlier and P. M. Ajayan, *Phys. Rev. Lett.*, 2000, **84**, 1716–1719.
6 X. Li, Q. Wang and P. Jena, *J. Phys. Chem. Lett.*, 2017, **8**, 3234–3241.
7 W. Liu, J.-y. Liu, J. Xia, H.-q. Lin and M.-s. Miao, *Nanoscale*, 2018, **10**, 11328–11334.
8 D. Malko, C. Neiss, F. Viñes and A. Görling, *Phys. Rev. Lett.*, 2012, **108**, 086804.
9 Z. Wang, X. F. Zhou, X. Zhang, Q. Zhu, H. Dong, M. Zhao and A. R. Oganov, *Nano Lett.*, 2015, **15**, 6182–6186.
10 L. Z. Zhang, Z. F. Wang, Z. M. Wang, S. X. Du, H. J. Gao and F. Liu, *J. Phys. Chem. Lett.*, 2015, **6**, 2959–2962.
11 Y. Liu, G. Wang, Q. Huang, L. Guo and X. Chen, *Phys. Rev. Lett.*, 2012, **108**, 225505.
12 X. Luo, L.-M. Liu, Z. Hu, W.-H. Wang, W.-X. Song, F. Li, S.-J. Zhao, H. Liu, H.-T. Wang and Y. Tian, *J. Phys. Chem. Lett.*, 2012, **3**, 3373–3378.
13 S. Zhang, J. Zhou, Q. Wang, X. Chen, Y. Kawazoe and P. Jena, *Proc. Natl. Acad. Sci. U. S. A.*, 2015, **112**, 2372.
14 M. Mina and O. Susumu, *Appl. Phys. Express*, 2013, **6**, 095101.
15 Y. Aierken, O. Leenaerts and F. M. Peeters, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2016, **94**, 155410.
16 M. Maruyama and S. Okada, *Diamond Relat. Mater.*, 2018, **81**, 103–107.
17 B. Ram and H. Mizuseki, *Carbon*, 2018, **137**, 266–273.
18 C. Zhang, F. Q. Wang, J. Yu, S. Gong, X. Li and Q. Sun, *Phys. Chem. Chem. Phys.*, 2018, **20**, 9123–9129.
19 M. Long, L. Tang, D. Wang, Y. Li and Z. Shuai, *ACS Nano*, 2011, **5**, 2593–2600.
20 S. Gao, H. Xiang, B. Xu, Y. D. Xia, J. Yin and Z. G. Liu, *Chin. Phys. Lett.*, 2016, **33**, 083101.
21 R. Krishnan, W.-S. Su and H.-T. Chen, *Carbon*, 2017, **114**, 465–472.
22 Y. Shen, F. Q. Wang, J. Liu, Y. Guo, X. Li, G. Qin, M. Hu and Q. Wang, *Nanoscale*, 2018, **10**, 6099–6104.
23 Y. Ma, H. Chang, M. Zhang and Y. Chen, *Adv. Mater.*, 2015, **27**, 5296–5308.
24 Y. Jiao, A. Du, M. Hankel, Z. Zhu, V. Rudolph and S. C. Smith, *Chem. Commun.*, 2011, **47**, 11843–11845.
25 G. Li, Y. Li, H. Liu, Y. Guo, Y. Li and D. Zhu, *Chem. Commun.*, 2010, **46**, 3256–3258.
26 R. H. Baughman, J. M. Shacklette, A. A. Zakhidov and S. Stafström, *Nature*, 1998, **392**, 362–365.
27 M. Schenk and S. D. Guest, *Proc. Natl. Acad. Sci. U. S. A.*, 2013, **110**, 3276.
28 K. E. Evans, M. A. Nkansah, I. J. Hutchinson and S. C. Rogers, *Nature*, 1991, **353**, 124–124.
29 L. Yu, Q. Yan and A. Ruzsinszky, *Nat. Commun.*, 2017, **8**, 15224.
30 Z. Qin, G. Qin and M. Hu, *Nanoscale*, 2018, **10**, 10365–10370.
31 G. N. Greaves, A. L. Greer, R. S. Lakes and T. Rouxel, *Nat. Mater.*, 2011, **10**, 823–837.
32 C. Huang and L. Chen, *Adv. Mater.*, 2016, **28**, 8079–8096.
33 K. E. Evans and A. Alderson, *Adv. Mater.*, 2000, **12**, 617–628.
34 R. Lakes, *Science*, 1987, **235**, 1038–1041.
35 B. D. Caddock and K. E. Evans, *J. Phys. D: Appl. Phys.*, 1989, **22**, 1877–1882.
36 K. E. Evans and B. D. Caddock, *J. Phys. D: Appl. Phys.*, 1989, **22**, 1883–1887.
37 J. Dagdelen, J. Montoya, M. de Jong and K. Persson, *Nat. Commun.*, 2017, **8**, 323.
38 J.-W. Jiang, T. Chang, X. Guo and H. S. Park, *Nano Lett.*, 2016, **16**, 5286–5290.
39 L. J. Gibson and M. F. Ashby, *Cellular Solids: Structure & Properties*, Cambridge University Press, New York, NY, 2nd edn, 1997.
40 J. N. Grima, L. Oliveri, D. Attard, B. Ellul, R. Gatt, G. Cicala and G. Recca, *Adv. Eng. Mater.*, 2010, **12**, 855–862.
41 X. Gong, J. Huang, F. Scarpa, Y. Liu and J. Leng, *Compos. Struct.*, 2015, **134**, 384–392.
42 L. J. Gibson, K. E. Easterling and M. F. Ashby, *Proc. R. Soc. London Ser. A*, 1981, **377**, 99–117.
43 Y. Wu, N. Yi, L. Huang, T. Zhang, S. Fang, H. Chang, N. Li, J. Oh, J. A. Lee, M. Kozlov, A. C. Chipara, H. Terrones, P. Xiao, G. Long, Y. Huang, F. Zhang, L. Zhang, X. Lepró, C. Haines, M. D. Lima, N. P. Lopez, L. P. Rajukumar, A. L. Elias, S. Feng, S. J. Kim, N. T. Narayanan, P. M. Ajayan, M. Terrones, A. Aliev, P. Chu, Z. Zhang, R. H. Baughman and Y. Chen, *Nat. Commun.*, 2015, **6**, 6141.
44 A. J. Mannix, X.-F. Zhou, B. Kiraly, J. D. Wood, D. Alducin, B. D. Myers, X. Liu, B. L. Fisher, U. Santiago, J. R. Guest, M. J. Yacaman, A. Ponce, A. R. Oganov, M. C. Hersam and N. P. Guisinger, *Science*, 2015, **350**, 1513.
45 Z. Gao, X. Dong, N. Li and J. Ren, *Nano Lett.*, 2017, **17**, 772–777.

46 R. Peng, Y. Ma, Z. He, B. Huang, L. Kou and Y. Dai, *Nano Lett.*, 2019, **19**, 1227–1233.
47 J.-W. Jiang and H. S. Park, *Nat. Commun.*, 2014, **5**, 4727.
48 H. Wang, X. Li, P. Li and J. Yang, *Nanoscale*, 2017, **9**, 850–855.
49 X. Wei, B. Fragneaud, C. A. Marianetti and J. W. Kysar, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2009, **80**, 205407.
50 G. Kresse and J. Furthmüller, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 1996, **54**, 11169–11186.
51 P. E. Blöchl, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 1994, **50**, 17953–17979.
52 J. P. Perdew, K. Burke and M. Ernzerhof, *Phys. Rev. Lett.*, 1996, **77**, 3865–3868.
53 J. Heyd, G. E. Scuseria and M. Ernzerhof, *J. Phys. Chem. C*, 2003, **118**, 8207–8215.
54 J. Heyd, G. E. Scuseria and M. Ernzerhof, *J. Chem. Phys.*, 2006, **124**, 219906.
55 G. Onida, L. Reining and A. Rubio, *Rev. Mod. Phys.*, 2002, **74**, 601–659.
56 A. Togo and I. Tanaka, *Scr. Mater.*, 2015, **108**, 1–5.
57 S. Eldin, J. F. Liebman, L. D. Reynolds and P. Dowd, *Tetrahedron Lett.*, 1992, **33**, 4525–4528.
58 V. Lavallo, Y. Canac, B. Donnadieu, W. W. Schoeller and G. Bertrand, *Science*, 2006, **312**, 722.
59 N. Narita, S. Nagai, S. Suzuki and K. Nakao, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 1998, **58**, 11009–11014.
60 L. Shi, P. Rohringer, K. Suenaga, Y. Niimi, J. Kotakoski, J. C. Meyer, H. Peterlik, M. Wanko, S. Cahangirov, A. Rubio, Z. J. Lapin, L. Novotny, P. Ayala and T. Pichler, *Nat. Mater.*, 2016, **15**, 634–639.
61 Y. Ding and Y. Wang, *J. Phys. Chem. C*, 2013, **117**, 18266–18278.
62 Y. Yang and X. Xu, *Comput. Mater. Sci.*, 2012, **61**, 83–88.
63 Z. Wang, F. Dong, B. Shen, R. J. Zhang, Y. X. Zheng, L. Y. Chen, S. Y. Wang, C. Z. Wang, K. M. Ho, Y.-J. Fan, B.-Y. Jin and W.-S. Su, *Carbon*, 2016, **101**, 77–85.
64 J. Isberg, J. Hammersberg, E. Johansson, T. Wikström, D. J. Twitchen, A. J. Whitehead, S. E. Coe and G. A. Scarsbrook, *Science*, 2002, **297**, 1670–1672.
65 J.-H. Choi, P. Cui, H. Lan and Z. Zhang, *Phys. Rev. Lett.*, 2015, **115**, 066403.
66 M. Zhang, L.-Y. Huang, X. Zhang and G. Lu, *Phys. Rev. Lett.*, 2017, **118**, 209701.
67 X. Wang, A. M. Jones, K. L. Seyler, V. Tran, Y. Jia, H. Zhao, H. Wang, L. Yang, X. Xu and F. Xia, *Nat. Nanotechnol.*, 2015, **10**, 517–521.
68 J. Bardeen and W. Shockley, *Phys. Rev.*, 1950, **80**, 72–80.
69 J. Xi, M. Long, L. Tang, D. Wang and Z. Shuai, *Nanoscale*, 2012, **4**, 4348–4369.
70 J. Qiao, X. Kong, Z. X. Hu, F. Yang and W. Ji, *Nat. Commun.*, 2014, **5**, 4475.
71 J. Chen, J. Xi, D. Wang and Z. Shuai, *J. Phys. Chem. Lett.*, 2013, **4**, 1443–1448.
72 L. Li, Y. Yu, G. J. Ye, Q. Ge, X. Ou, H. Wu, D. Feng, X. H. Chen and Y. Zhang, *Nat. Nanotechnol.*, 2014, **9**, 372–377.
73 L. Brus, *J. Phys. Chem.*, 1994, **98**, 3575–3581.
74 S. Grimme, J. Antony, S. Ehrlich and H. Krieg, *J. Chem. Phys.*, 2010, **132**, 154104.

This journal is © The Royal Society of Chemistry 2020